

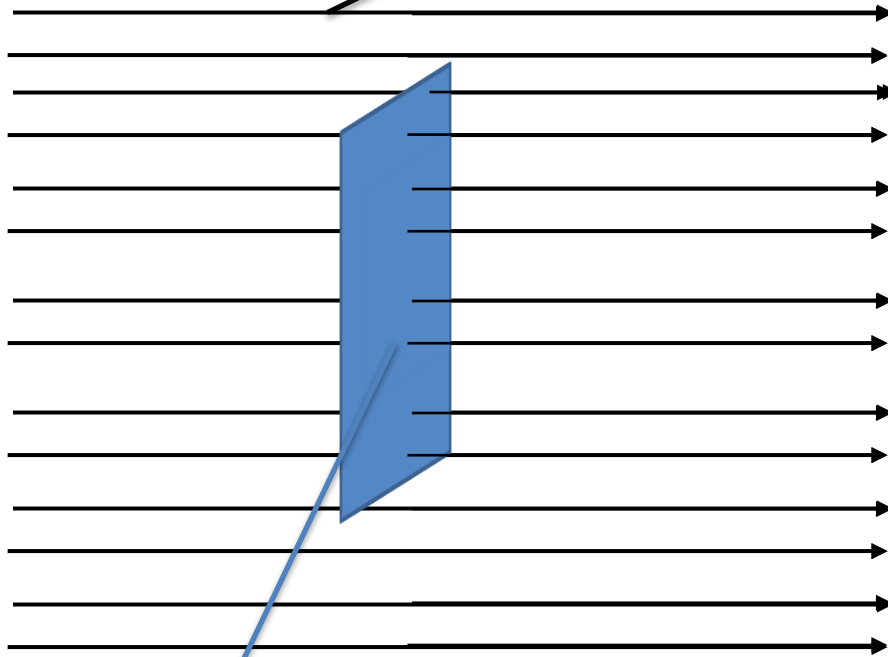


LESSON 3

- BOOK CHAPTER 23
- GAUSS' LAW

Electric flux

Uniform Electric Field



$$2\vec{E}$$

Look at the no. of field lines of field strength $\vec{E}$ penetrating the surface.

What happens if we increase the field strength to $2\vec{E}$?

What if we increase the surface area to $2A$?

Vertical Surface placed in Electric field.

Electric Flux:

The electric flux Φ through a surface is the amount of electric field that pierces the surface.

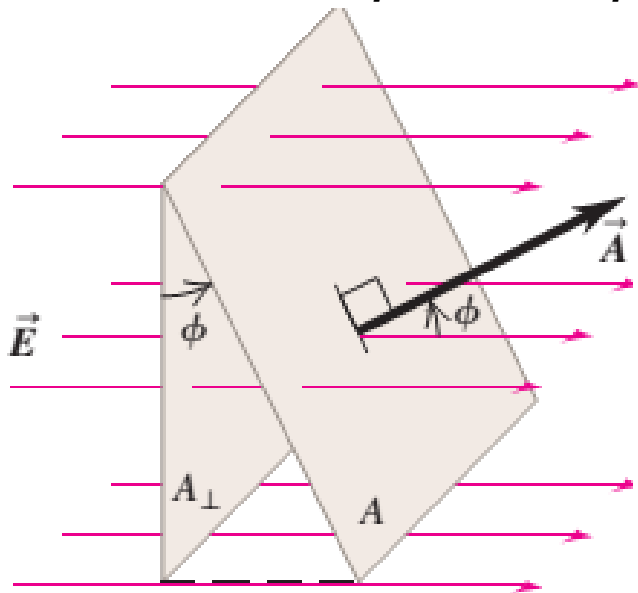
Electric Flux for uniform $\vec{E}$ and through a flat surface (open plane):

The electric flux ϕ through the flat surface equals the scalar product of the electric field $\vec{E}$ and the area vector $\vec{A}$.

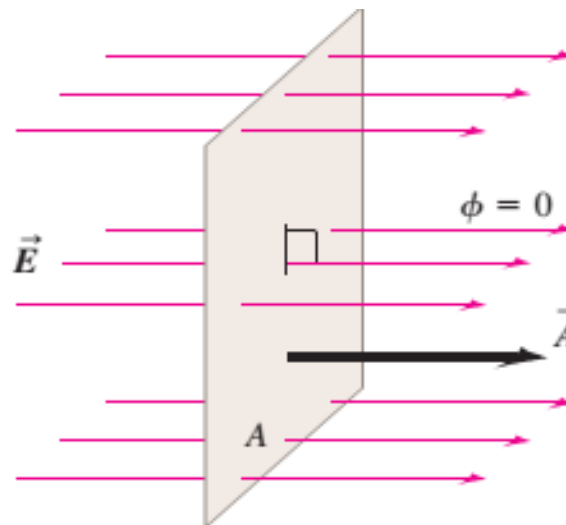
That is, $\Phi = \vec{E} \cdot \vec{A}$

We can represent the direction of a vector area $\vec{A}$ by using a unit vector $\hat{n}$ perpendicular to the area: $\hat{n}$ stands for normal." Then $\vec{A} = A\hat{n}$

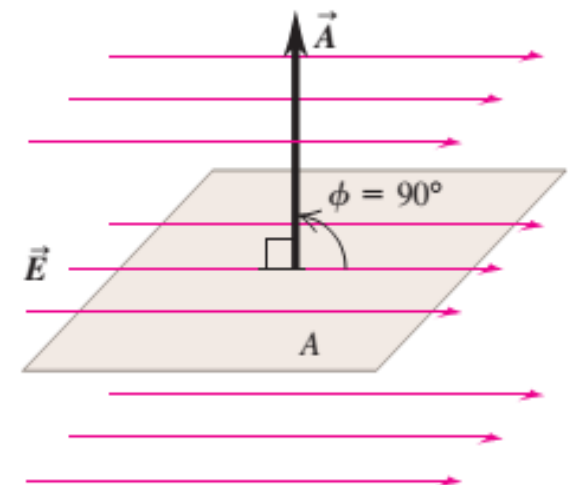
We can write as $\phi = EA \cos \phi$



$$\phi = EA \cos 0^\circ = EA$$



$$\phi = EA \cos 90^\circ = 0 \frac{N \cdot m^2}{C}$$



Electric Flux for nonuniform $\vec{E}$ and through a curved surface:

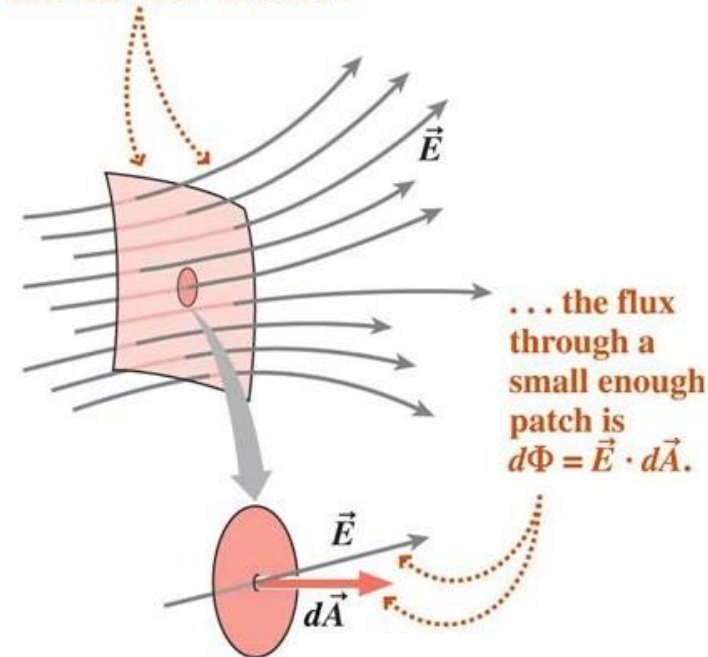
When the surface is curved and field is nonuniform, we calculate the flux by dividing the surface into small (differential) patches (areas) dA , so that each differential area is essentially flat and field is uniform over each patch.

The total flux through the surface is given by

$$\Phi = \int \vec{E} \cdot d\vec{A}$$

where the integration is carried out over the surface.

Although the surface curves and the field varies ...



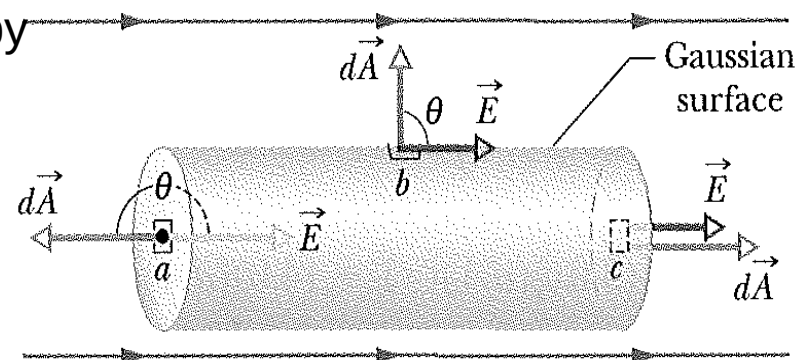
Electric Flux through a closed surface:

The net flux through a closed surface is given by

$$\Phi = \oint \vec{E} \cdot d\vec{A}$$

$$\Phi = \int \vec{E} \cdot d\vec{A} + \int \vec{E} \cdot d\vec{A} + \int \vec{E} \cdot d\vec{A}$$

where the integration is carried out over the entire surface.



Gauss' law:

Gauss's law is an alternative to Coulomb's law. While completely equivalent to Coulomb's law, Gauss's law provides a different way to express the relationship between electric charge and electric field. It was formulated by Carl Friedrich Gauss(1777–1855), one of the greatest mathematicians of all time

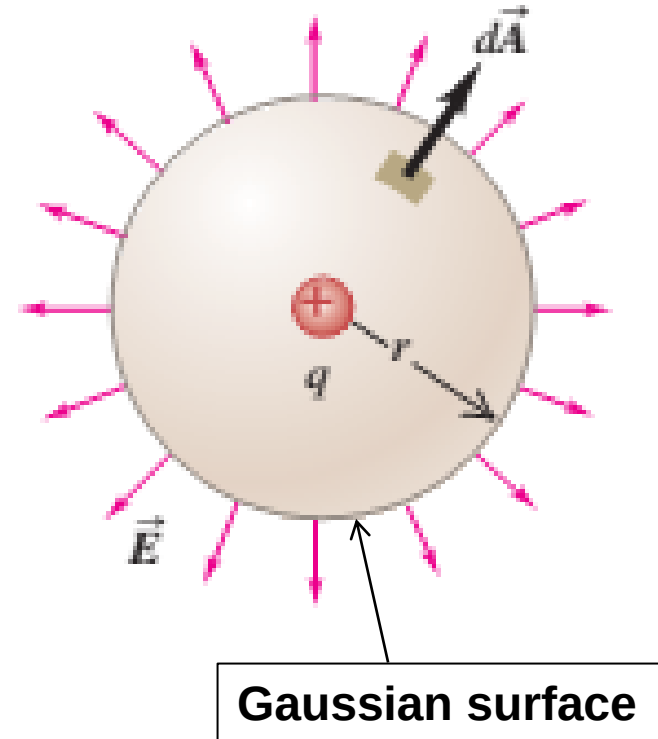
Gauss's law states that the total electric flux(Φ) through any closed surface, known as Gaussian surface, (a surface enclosing a definite volume) is proportional to the total (net) electric charge(q) inside the surface. That is

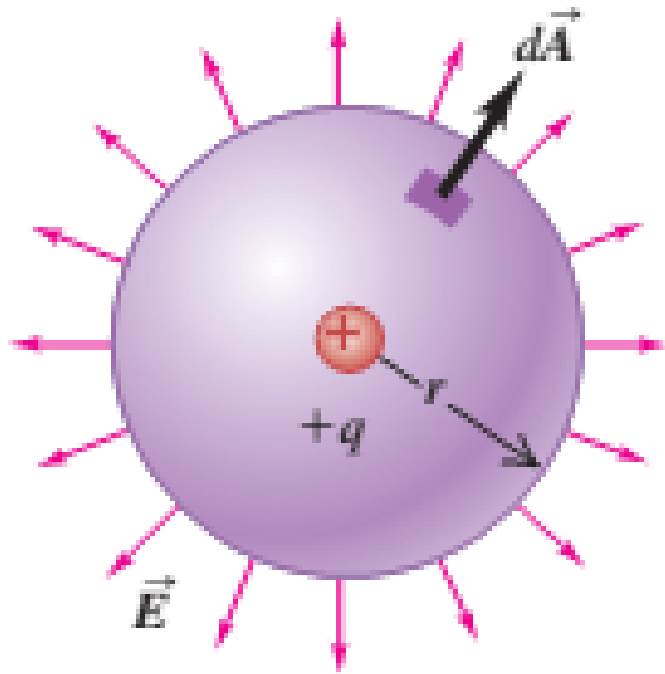
$$\Phi = \frac{q}{\epsilon_0}$$

We can write the above form as

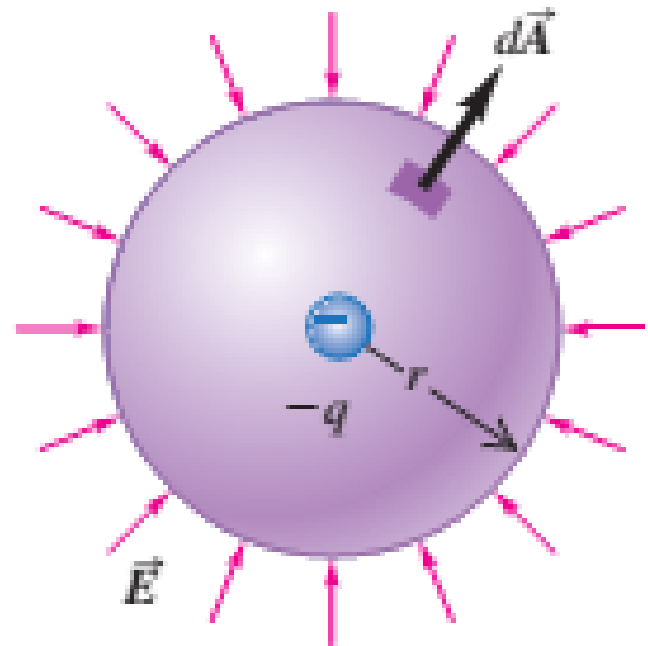
$$\epsilon_0 \Phi = q$$

$$\epsilon_0 \oint \vec{E} \cdot d\vec{A} = q$$





Gaussian surface around positive charge:
positive (outward) flux



Gaussian surface around negative charge:
negative (inward) flux

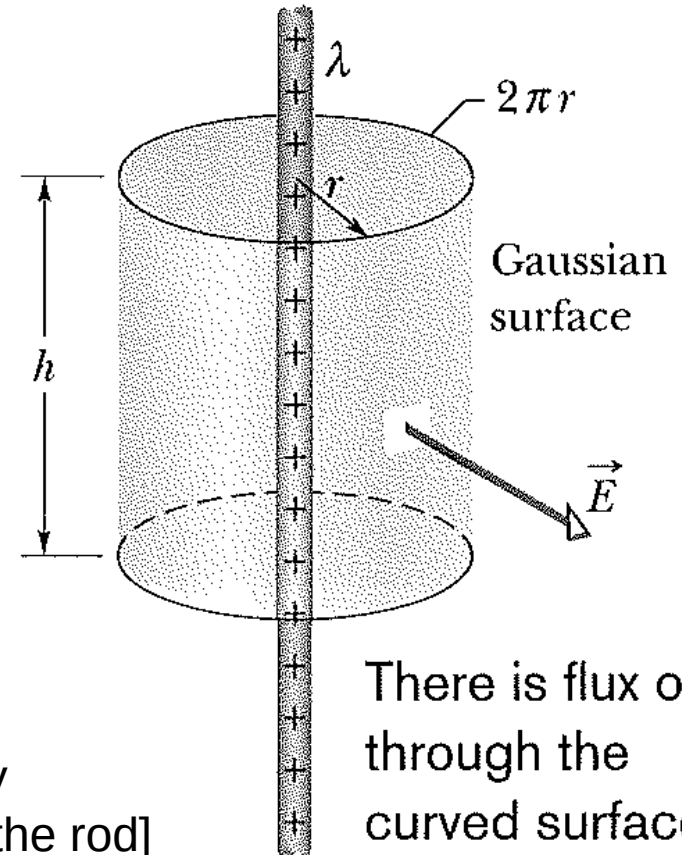
Applying Gauss' Law: Cylindrical Symmetry:

An infinitely long cylindrical plastic rod with a uniform charge density λ

We choose a circular cylinder of radius r and length h , coaxial with the rod.

Because the Gaussian surface must be closed, we include two end caps as part of the surface.

At every point on the cylindrical part of the Gaussian surface, $\vec{E}$ must have the same magnitude E and (for a positively charged rod) must be directed radially outward.



Using Gauss' law, we can write

$$\epsilon_0 \Phi = q = \lambda h \quad [\lambda \text{ is the liner charge density (charge per unit length) of the rod}]$$

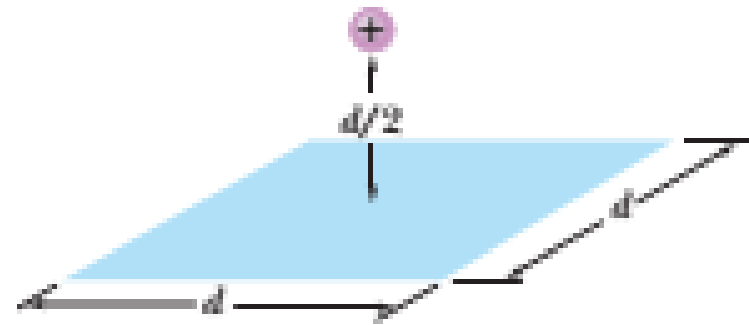
$$\epsilon_0 EA = \lambda h \quad [\text{There is no flux through the end caps because the direction of } \vec{E} \text{ is radially outward from the line of charge.}]$$

$$\epsilon_0 E(2\pi r h) = \lambda h \quad \text{Finally,}$$

$$E = \frac{\lambda}{2\pi\epsilon_0 r}$$

Problem 5 (book chapter 23)

A proton is a distance $d/2$ directly above the center of a square of side d . What is the magnitude of the electric flux through the square? (*Hint: Think of the square as one face of a cube with edge d .*)



Answer:

Consider a cube consisting with six square faces; each of edge d . Thus, the proton is enclosed by the cubical Gaussian surface.

Electric flux through the cubical Gaussian surface is

$$\Phi = \frac{q}{\epsilon_0} = \frac{e^+}{\epsilon_0} = \frac{1.6 \times 10^{-19}}{8.854 \times 10^{-12}} = 0.18 \times 10^{-7} = 18 \times 10^{-9} \frac{N.m^2}{C}$$

Electric flux through a face (a square) of a cube is

$$\Phi_s = \frac{18 \times 10^{-9}}{6} = 3 \times 10^{-9} \frac{N.m^2}{C}$$

Problem 25 (Book chapter 23)

An infinite line of charge produces a field of magnitude $4.5 \times 10^4 \text{ N/C}$ at distance 2.0 m. Find the linear charge density.

Answer:

Required formula:

$$E = \frac{\lambda}{2\pi\epsilon_0 r}$$

$$4.5 \times 10^4 = \frac{2\lambda}{4\pi\epsilon_0 r} = 9 \times 10^9 \times 2\lambda/2$$

$$\lambda = \frac{4.5 \times 10^4}{9 \times 10^9} = 0.5 \times 10^{-5} = 5 \times 10^{-6} \text{ C/m}$$

Given,

$$E = 4.5 \times 10^4 \text{ N/C}$$

$$r = 2.0 \text{ m}$$

$$\lambda = ?$$

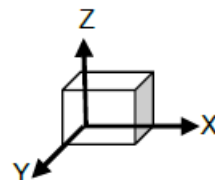
Let's try !!!

A. Calculation of Electric flux:

1. A surface has the area vector $\vec{A} = (2\hat{i} + 3\hat{j})\text{m}^2$. What is the flux of an electric field through it, if the field is (a) $\vec{E} = 4\hat{i}\text{ N/C}$ and (b) $\vec{E} = 5\hat{k}\text{ N/C}$

B. Calculation of Electric flux from Gauss's Law

2. A point charge of $1.8\text{ }\mu\text{C}$ is placed at the center of a cubical Gaussian surface 55 cm on edge. What is the electric flux through the surface?
3. A point charge causes an electric flux of $-750\text{ N m}^2/\text{C}$ to pass through a spherical Gaussian surface of 10.0 cm radius centered on the charge. (a) If the radius of the Gaussian surface were doubled, how much flux would pass through the surface. (b) What is the value of the point charge?
4. A point charge $q_1 = 5.0\text{ nC}$ is located at the origin, and a second point charge $q_2 = -3.0\text{ nC}$ is on the x axis at $x = 1.0\text{ m}$. What is the total electric flux due to these two point charges through a spherical surface centered at the origin and with radius a) 0.5 m, b) 1.5 m, and c) 2.5 m?



THANK YOU